Chimata Sai Nihal

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EDUCATION

Manipal Institute of Technology, Bangalore **Aug 2024 – Present**
B.Tech, Electronics and Computer Engineering (ECM) with Honours CGPA: 8.55 / 10

TECHNICAL SKILLS

Languages: C, C++, Java, Python, Bash, HTML, CSS, JavaScript
Frameworks & Tools: OpenCV, MediaPipe, Pygame, LangChain (working knowledge), Git, GitHub, Arduino IDE, Docker
Hardware: Arduino, IR/ultrasonic sensors, motor drivers and actuators; breadboard prototyping; CRO, AFO, DMM
Interests: Full Stack Development, Backend Systems, System Design, Cloud & DevOps, APIs & Microservices, Applied ML, Computer Vision

PROJECTS

Incident Response Agent
Python, LangChain, LLM Agents, REST APIs | github.com/sainihal-chimata/incident-response-agent
- Built an AI-based incident response system that monitors incoming alerts, classifies severity, correlates logs across services, and suggests remediation actions through an LLM pipeline.
- Implements iterative decision logic to escalate, query additional context, or propose a fix; integrated with REST APIs for alert ingestion and structured JSON outputs. Fulfilled all Round 1 criteria — Scaler x Meta Hackathon. Results awaited.

Gesture-Based Air Drawing
Python, OpenCV, MediaPipe | github.com/sainihal-chimata/Gesture-Tracking-Air-Drawing-System
- Developed a real-time gesture-controlled drawing system using hand landmark detection and motion tracking; strokes are rendered on a virtual canvas overlaid on the live webcam feed.
- Supports multi-color drawing, erase mode, and canvas clear via keyboard shortcuts; base system fully functional with gesture-based mode switching and additional gestures planned.

Drone Path Optimization
Python, Pygame, Graph Algorithms | github.com/sainihal-chimata/Path-Finding-Drone-Simulation
- Designed a path optimization system modelling delivery waypoints as a weighted directed graph; uses DFS to enumerate all routes and a composite cost function (distance, energy, time, penalty with tunable $\alpha/\beta/\gamma$ weights) to select the globally optimal path.
- Built a Pygame visualization that animates drone traversal along every enumerated path and highlights the best one in real time.

Wireless Tank Robot with Robotic Arm Integration
Arduino, HC-05, Motor Drivers, Servo Motors | Robot Independent
- Designed and assembled a tracked robot with wireless motor control over Bluetooth.
- Focused on embedded C programming, motor driver logic, and the mechanical and software integration of arm kinematics into the existing platform.

Autonomous Navigation Simulation
ROS, Gazebo, RViz | Simulation Demo
- Completed an autonomous navigation and obstacle avoidance simulation in Gazebo/RViz, implementing path planning and obstacle detection for a wheeled robot.

Robotics Workshop Projects — ART-PARK, IISc Bangalore
Arduino, IR/Ultrasonic Sensors, Embedded C ***Certified***
- Independently enrolled in an intensive hands-on robotics workshop; built line-following, obstacle-avoidance, and Bluetooth-controlled wheeled robots with emphasis on sensor interfacing, motor control, and embedded programming.
- Awarded a completion certificate by ART-PARK, IISc Bangalore upon successful project demonstration.

OPEN SOURCE

Google Summer of Code (GSoC) 2026 — Honeynet Project 2026
Python, Django, ReactJS, LangChain, Ollama, Docker | Project: IntelOwl
- Submitted a proposal to integrate a self-deployed LLM chatbot into IntelOwl (a threat intelligence platform) — proposed a RAG pipeline using LangChain, a locally hosted LLM via Ollama, a ReactJS chat sidebar, and a Django API backend. Selection results awaited.

ACTIVITIES & COMPETITIONS

- Scaler x Meta Hackathon 2025 — built an Incident Response Agent; project qualified Round 1 evaluation criteria. Final results awaited.
- College-level Hackathon — participated in initial rounds; gained experience in team-based problem solving under time constraints.
- MAVRO Autonomous Buggy Club, MIT Bangalore — reached finalist stage of the selection process.